Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/19/2013
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Antigen-Specific CD4+ T Cell Priming and Memory Response During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
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PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	04/15/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 ; ISS NOTE: Extended to 7/31/2014 per A. Chu/ARC and NSSC information (Ed., 4/3/2014) NOTE: New end date is 4/13/2014 per A. Chu/ARC and NSSC (Ed., 2/21/2013) NOTE: New end date is 4/13/2013 per A. Chu/ARC (Ed., 1/27/2012) NOTE: New end date is 4/14/2012 per A. Chu/ARC (Ed., 3/10/2011) NOTE: Grant number corrected to NNX09AH21G (from NNX09AG21G) per A. Chu/ARC (4/2010) NOTE: End date changed to 4/14/2011 (from 4/14/2012) per A. Chu/ARC (7/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AH21G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Spaceflight immunosuppression is a significant obstacle to long-term human space travel. Of foremost concern is whether astronauts/cosmonauts may be able to generate effective protective immune responses against infections while in spaceflight. Using an innovative mouse experimental model, we propose a set of experiments that will test 1) whether initial specific activation of T cells is intact and 2) whether memory T cell function is maintained during spaceflight. Memory T cells are the mediators of immune protection after vaccinations and whether they function properly during spaceflight will determine if pre-flight vaccinations may be useful interventions. In our experimental system, transgenic T cells specific for a model protein will be transferred into recipient mice that have implanted mini-pumps. The mini-pumps will release the model protein along with non-infectious purified bacterial cell wall components. Specific T cells stimulated under these inflammatory conditions will be similar to those activated in a bacterial infection but no infectious agents will be involved. Mini-pumps will delay T cell stimulation until after mice acclimate to spaceflight to ensure that initiation of T cell activation occurs in microgravity. Naïve (previously unactivated) T cells will be transferred to determine whether responses to new immunological challenges (e.g. new infections) are intact. Previously activated T cells that will develop into memory cells will be transferred to determine whether robust secondary responses may be elicited during spaceflight. Our proposal directly addresses the research priorities of the Bion-M1 mission (changed to a Space Shuttle mission in 2010) to investigate resistance to infections and acquired immune responses without using infectious agents onboard the spacecraft. Together, these proposed experiments will address some of the most pressing outstanding questions in spaceflight immunology and more clearly define the risks and potential countermeasures of spaceflight immunosuppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The alteration of the immune system in spaceflight has been suggested by some to have many analogies to changes in the immune system during aging. In these studies of the immune system in mice, we have found that the changes in the immune function in flown mice are much the same as in humans. This suggests that there are common mechanism of spaceflight immunosuppression in both mammals. Discoveries made from this study will help us understand the immune system and the basic initial steps of activation as well as help find new targets to use in the treatment of immunosuppression observed in aging and disease.
Task Progress:	Preparation of the manuscript is in progress for T-cell prime. In addition, further analysis of the data reveals that immune changes seen in the human are also present in the mice, suggesting a common mechanism of action of microgravity alteration in mammalian immune function. We were also able to augment our studies by examining the effects of hyperoxia (experience by astronauts during extra vehicular activity) on the immune system. The decrease of immune function by hyperoxia not only is important to EVA astronauts, but also to patients administered high levels of oxygen in the intensive care unit.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Abstracts for Journals and Proceedings	Hughes-Fulford M, Meissler J, Aguayo, E, Canderario TT, Globus R, Aguado J. "Hyperoxia Inhibits T cell Activation in Mice." 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012
Articles in Peer-reviewed Journals	Grenon SM, Saary J, Gray G, Vanderploeg JM, Hughes-Fulford M. "Can I take a space flight? Considerations for doctors." British Medical Journal. 2012 Dec 13;345:e8124. PMID: 23241271 , Dec-2012
Articles in Peer-reviewed Journals	Hughes-Fulford M, Meissler J, Aguayo ET, Globus R, Aguado J, Candelario T. "Hyperoxia inhibits T cell activation in mice." Journal of Gravitational Physiology 2012. Proceedings of 12th Annual European Life Sciences symposium, Aberdeen Scotland. In Press, as of December 2012. , Dec-2012